Highlights

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- We have integrated feature processing frameworks that utilize self-constructing graph (SCG) from multilevel hierarchical feature maps to generate graph representations.
- Feature refinement via Autoregressive moving average (ARMA) graph convolution was explored. IncARMAG is the first architecture to exploit both SCG and ARMA convolution for medical image classification.
- Based on evaluation on MedMNIST 2D datasets, IncARMAG achieved good performance that rivals deep learning models dedicated for medical image classification.
- IncARMAG has outperformed various CNN-, Transformer-, and hybrid-based models when benchmarked on similar datasets.

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Abstract

Effective computer-aided detection and diagnosis algorithms are being sought to carry out complex pattern recognition with great precision while increasing efficiency and robustness through automation. In this study, we leveraged self-constructing graph (SCG) and autoregressive moving average (ARMA) graph convolution to postprocess multi-level feature maps obtained from an Inception V3 network for medical image classification. The adopted SCG learns latent embeddings from the feature maps, which are subsequently used to construct graph topology based on embedding similarity. The ARMA filter then theoretically encodes more refined convolutional responses that simultaneously takes into account long-range dependencies from the SCG graph. The proposed IncARMAG model was evaluated on 2D MedMNIST datasets, which consist of several imaging modalities and classification tasks in the medical domain. Results from our experiments on MedMNIST datasets showed that IncARMAG outperformed state-of-the-art models on many cases. IncARMAG has also achieved the highest accuracy based on evaluation of various benchmark CNN, transformers, and hybrid models on four selected datasets. These results highlight how IncARMAG is a strong candidate in medical image analysis involving classification tasks.

Keywords: Medical image classification, Convolutional neural network, Graph convolution, Autoregressive moving average convolution, Feature embedding

1. Introduction

Medical imaging has been at the forefront of medical diagnosis and prognosis in several diseases, such as cancer and neurodegenerative diseases. Recent advancements in hardware paved the way to capture in vitro and in vivo environments and to reduce acquisition time, allowing fast evaluation and treatment delivery [1]. Radiographic x-ray, computed tomography (CT), fluorosence, and live-cell microscopy are some examples of imaging modalities that are widely used in the clinical domain. In addition to anatomical insights from images, physiological and other biophysical features can now be extracted with the imaging technology available today. Examples of imaging modality capable of such tasks are positron emission tomography (PET), magnetic resonance imaging (MRI), and fluorescence imaging [2].

Artificial intelligence (AI) in computer-aided diagnosis and medical interventions have gained an ever-increasing attention due to their capability to handle complex tasks with excellent performance. The complementary contributions of AI in medical imaging includes automation of detection, decision-making support, and image quality improvement [3]. The application of AI for automated disease localization and detection can lead to higher throughput of patients while simultaneously reducing physician-to-physician assessment variation [4]. Since many AI models excel in pattern recognition, these models can be used as data-driven mapping functions from images to a set of target outputs with great accuracy to aid clinical decision-making. This also maximizes information extraction from images compared to relying solely on visual inspection from experts [5].

Image classification/regression, segmentation, and object detection are some of the commonly applied tasks of AI in medical imaging. More advanced computer vision tasks, such as image reconstruction, registration, denoising, and generation, are currently becoming more relevant in today's medical imaging applications [6]. Machine learning algorithms coupled with feature engineering were the popular choice in the earlier days for carrying out medical image classification. In this case, handcrafted features, such as first-order, geometric, and texture features, are extracted and formed as vectorized input to a classical machine learning model [7]. Deep learning (DL) algorithms have been pivotal in image classification as they were shown in to achieve good performance in many instances. Artificial neural networks (ANNs), which are the foundation of DL algorithms, enable the features to be automatically learned and optimized to the task at hand [8].

Convolutional neural network (CNN) is a DL approach that has been widely used in image analysis, including image classification . CNN maps meaningful hierarchical features from imaging data through the subseqeuent application and optimization of cascaded convolutional filters [9, 10]. On the feature extraction part, the computational load does not scale with image size due to weight sharing of convolution operation. AlexNet [11] and VGG [12] were some of the pioneer models in the early image classification open-challenge, which instigated the rapid revolution of deep learning on computer vision. In recent years, tremendous improvement on CNN architecture has been made, emphasizing on accuracy, computational efficiency, and scalability at various domains [13]. For instance, a progressive increase in convolutional layer depth exhibited an initial increase in accuracy, followed by a decline in accuracy after a certain depth due to vanishing or exploding gradient problem during backpropagation. This problem was rectified in residual network (ResNet) models via identity mapping or shortcut connections [14]. Various CNN designs, such as EfficientNets, are based on neural architecture search methods that jointly optimize speed and accuracy [15, 16]. Squeeze-and-excitation, partial dense, and parallel multi-resolution are some examples of complex convolution blocks built to address various issues in CNN architectures [17, 18, 19].

Graph neural networks (GNNs) is a deep learning method dedicated for representation learning of graphs [10]. However, GNNs can be extended to regular grid-structured data by finding and constructing relational information among the spatial grids. Preprocessing steps of graph construction from images usually involve superpixel construction and/or prior application of convolutional feature extraction to form the graph topology [20, 21]. Graph convolution layers, which consists of message passing, aggregate, and update operations, are stacked to process and refine node and/or edge features. Different approaches on these operations is what makes up for the essence of various types of graph convolution. For example, graph attentional operator applies an attention mechanism during aggregation of graph features, in contrast to the equal weighting used in standard graph convolutional network (GCN) layer [22].

As it was shown in multiple studies on GNN-based vision tasks, there is a benefit in learning relation information among feature maps [21]. This study proposed a novel hybrid CNN-GNN model for image classification applied in the medical domain. The key contributions of this paper are as follows:

1. The model architecture proposed in this study utilized a multi-level feature map refinement with GNN. An Inception V3 backbone initially extracts features. Then, we adopted the self-constructing graph (SCG) framework to construct the graph topology at various feature map levels.
2. Instead of the conventional GCNs, we used autoregressive moving average (ARMA) graph convolutions for the GNN-based processing of the multi-level graph topology. Combining all the components, we henceforth refer to this architecture as IncARMAG.
3. Experiments to evaluate the performance of the developed IncARMAG model was conducted on multiple medical image classification datasets. Our proposed model was benchamrked against state-of-the art (SOTA) image classification models.
4. IncARMAG showed superior performance on most of the datasets compared to SOTA classification models.

2. Related Works

2.1. GNN Networks for Image Classification

As emphasized, graph representation can be obtained directly from images or feature maps. In the work of Monti et al. [23], learnable Gaussian kernels were applied to determine the connections based on geometric distances of superpixels on images. Using clustering methods, such as SLIC and SEED, it is possible to directly generate graph representations from images. Avelar et al., has used K-nearest neighbors (KNN) and SLIC clustering algorithm to generate the graph topology from the images. Graph attention network (GAT) layers was then used to encode for node embeddings from the input superpixel graphs [24]. A hybrid GNN architecture of GCN, GraphSAGE, and ARMA convolution filters following SLIC clustering was proposed by Yao et al. [25] for hyperspectral image classification.

Numerous GNN model designs for vision tasks incorporates CNN or transformer architecture to optimize deep feature extraction. Contextual information from a series of images was modelled through a combination of CNN and GNN in the work of Campos et al. [26]. Node features and edges were calculated based on the initial CNN feature maps of bounding boxes, followed by subgraph integration and subsequent GNN layer operations. VisionG (ViG) stands as one of the most successful application of GNN models in image classification. ViG introduced the grapher module, which better alleviates over-smoothing compared to the standard GCN layer by adding nonlinear activation, residual connection, and MLP layer [27]. Based on their benchmarking experiments on ImageNet dataset, ViG has outperformed representative CNN, MLP, and transformer models. In the study of Fei et al. [28], Euclidean convolution and GCN modules were arranged in parallel to simultaneously capture local and global semantic features.

2.2. Medical Image Classification

As vision models continue to advance, their ongoing application in the medical field becomes increasingly prevalent. Classification algorithms that were originally developed and/or benchmarked from other domains are continuously being adopted in the medical field. For example, the ResNet models have been used in numerous medical imaging classification tasks, such as cervical cancer identification [29], breast cancer detection [30], and pulmonary diseases [31]. Other models and learning methods have played prominent roles in advancing medical image classification. This includes developing and/or integrating multiple architectural modules. Yengec-Tasdemir et al. [32] employed contrastive learning to improve the performance of various CNN models for polyp classification.

Another approach to vision task is by the application of transformers. Transformers was adopted from natural language processing (NLP) for image classification, wherein an input image is divided into patches and treated as tokens prior to feature embedding and processing. MedViT is a hybrid CNN-transformer model that combines the superior inductive bias of convolution, while gaining the global receptive field from attention mechanism [33]. Their evaluation on MedMNIST datasets, a repository of 2D and 3D image datasets from different medical modalities [34], showed a superior average accuracy of MedViT over ResNet, auto-sklearn, AutoKeras, and Google AutoML. Similarly, MedMamba models both global relations within the multidirectional input images or feature maps via a multi-branch model of convolution and structured state-space sequence model [35]. This state-space sequence model is based on an iterative solution of a linear ordinary differential equation (ODE), which can also be generalized into a single 1-D global convolution. In most cases, MEDMAMBA achieved higher performance on most of the MedMNIST datasets compared to MedViT.

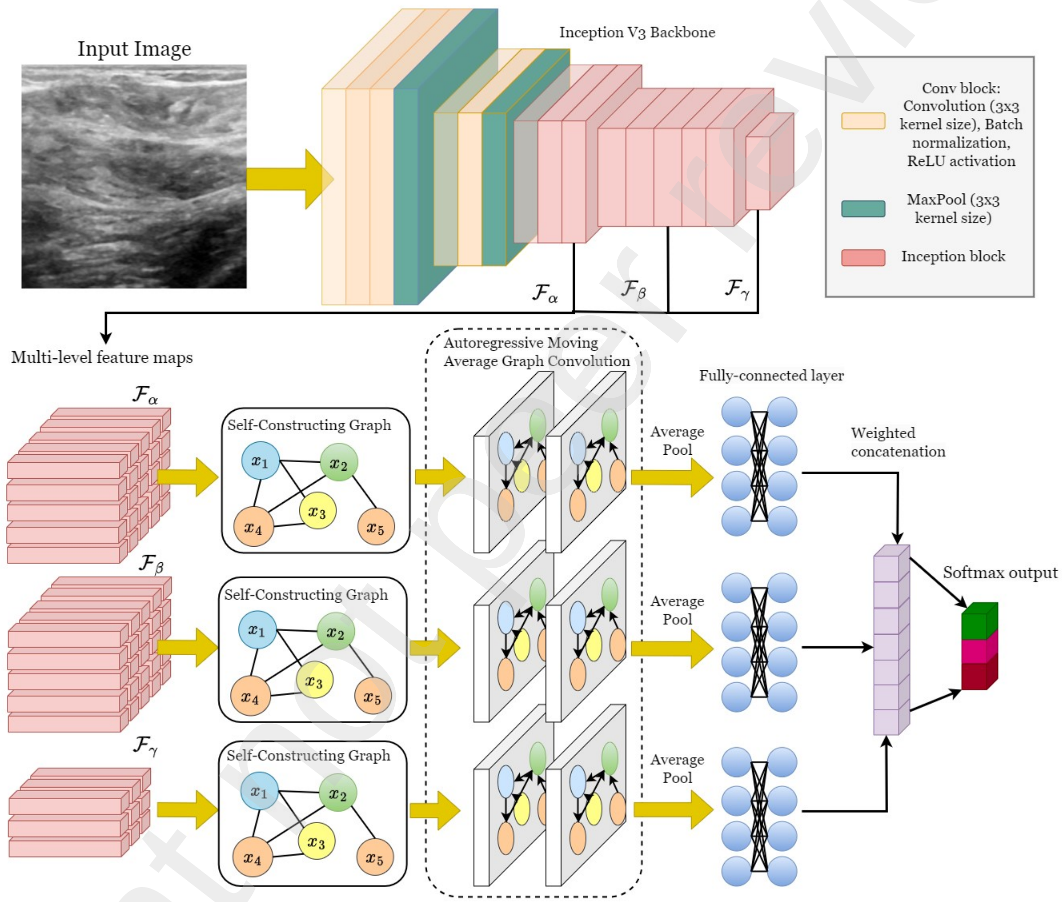

Figure 1: The proposed IncARMAG architecture, consisting of the following stages: (1) an Inception V3 backbone, (2) Self-constructing graph module, (3) ARMA GNN processing, and (4) weighted concatenation and classifier head.

3. Methods

An overview of the proposed IncARMAG architecture is illustrated in Fig. 1. Details of each component of the IncARMAG model will be discussed in the following subsections.

3.1. Deep Feature Extraction

A motivation of the original Inception architecture is to capture the correlation statistics of feature maps at various scales, wherein the kernel size reflects the spread of the correlated cluster units [36]. Another advantage of the Inception block is in the reduction of computational cost via depthwise filter concatenation and dimensionality reduction. In the improved Inception V3 model, computational cost was further reduced by factorizing the 5x5 convolutional kernel into a series of smaller and spatially asymmetric kernels [37].

Our architecture used the Inception V3 architecture, omitting the final two Inception blocks and the fully-connected head classifiers. Reducing computational cost was the primary reason behind the omission of these layers. It was also argued that adding auxiliary classifiers that branches from low-level inception blocks slightly boost model performance, which might be caused by either the refinement of low-level features or the regularization by the auxiliary heads [38]. This has inspired us to investigate the effect of replacing simple MLP layers with GNN as auxiliary classifiers. Our IncARMAG architecture has three GNN processing branches that correspondingly take three feature maps from different stages of the Inception V3 backbone. We denote these feature maps as $\mathcal{F}_\alpha$, $\mathcal{F}_\beta$, and $\mathcal{F}_\gamma$. $\mathcal{F}_\gamma \in \mathbb{R}^{7\times7\times2048}$ corresponds to the feature map of the last Inception block, whereas $\mathcal{F}_\alpha$ and $\mathcal{F}_\beta$ are from the preceding stages of the backbone network. Two variants of IncARMAG model with different depth locations of $\mathcal{F}_\alpha$ and $\mathcal{F}_\beta$ were developed in this study.

3.2. Self-constructing graph topology

Prior to graph convolution processing of extracted deep features, the feature maps must be converted to their representative graph topology. In general, a graph can be defined by an ordered pair, $G = \{\mathcal{N}, \mathcal{E}\}$, where $\mathcal{N}$ and $\mathcal{E}$ are the set of nodes and edges, respectively. The graph topology can also be represented via the adjacency matrix, $A \in \mathbb{R}^{|\mathcal{N}|\times|\mathcal{N}|}$, where $|\mathcal{N}|$ is the

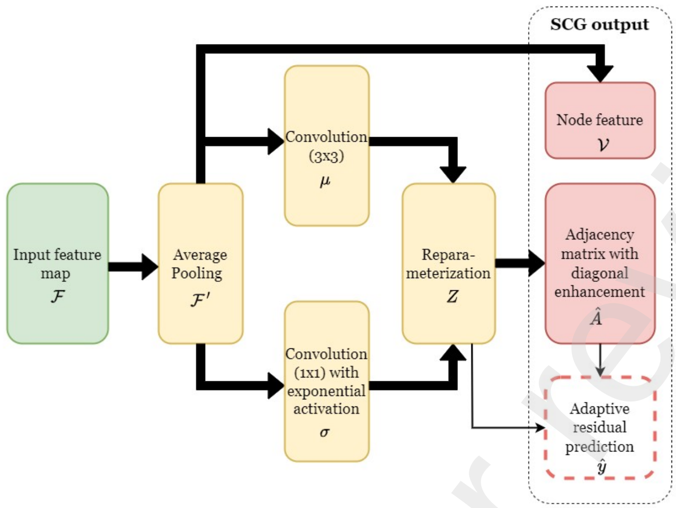

Figure 2: The operations involved in the SCG algorithm. SCG generates a graph, defined by a node feature and adjacency matrix, as the output. In addition, residual prediction from SCG are also added on final GNN layers.

number of nodes. The SCG algorithm, proposed by Liu et al. [39], was employed to automatically learn the adjacency matrix given the input feature map.

Fig. 2 provides a schematic description of the SCG module, as adopted from [39]. For any feature map, denoted simply as $\mathcal{F}$, an adaptive average pooling was used to reduce the dimensions of the feature map from $\mathcal{F} \in \mathbb{R}^{n\times n\times c}$ to $\mathcal{F}' \in \mathbb{R}^{n'\times n'\times c}$, where c is the number of convolutional filters while n and n' are the feature dimensions before and after pooling, respectively. The pooled features are considered as node features in the graph, wherein each node has a node feature equal to depth-wise channel vector of a particular 2D location in the feature map. This is equivalent to reshaping the pooled feature map: $\mathcal{F}' \in \mathbb{R}^{n'\times n'\times c} \to \mathbb{R}^{N\times c}$, where $N = |\mathcal{N}| = n' \times n'$.

Generating the adjacency graph via SCG requires embedding the feature map into a latent space, parameterized by mean, μ, and standard deviation, σ matrices. The mean matrix, μ, is obtained from the pooled features using a convolution with 3×3 kernel size, whereas a 1×1 convolution with exponential activation is applied to obtain the σ matrix. These matrices encodes a latent embedding, expressed in the form:

$$Z = \mu + \sigma \cdot \mathcal{N}(0, I) \rightarrow Conv_{3\times3}(\mathcal{F}') + \exp[Conv_{1\times1}(\mathcal{F}')] \cdot \mathcal{N}(0, I) \tag{1}$$

where $\mathcal{N}(0, I)$ is a Gaussian distributed noise present during training time. Connection between two nodes is established if their embedding projection is nonnegative, which means that the initial graph adjacency matrix is given by:

$$A = ReLU\left(ZZ^T\right) \tag{2}$$

where $ReLU$ is the rectified linear unit activation function [40].

Compared to the original graph auto-encoders, SCG introduced diagonal enhancement through the adaptive factor:

$$\gamma = \sqrt{1 + \frac{N}{\sum_{i=1}^{N} A_{ii} + \epsilon}} \tag{3}$$

where A_{ii} is the diagonal entries of the adjacency matrix, and ϵ is a smoothing factor. Combining the diagonal enhancement with the adjacency matrix normalization, the output adjacency matrix from SCG is:

$$\hat{A} = D^{-\frac{1}{2}} \left(A + \gamma \cdot \mathrm{diag}(A) + I\right) D^{\frac{1}{2}} \tag{4}$$

where D is the degree matrix with the diagonal entries equal to the degree of the corresponding node.

In addition learning the graph from the feature maps, a residual prediction term, given by:

$$\hat{y} = \gamma \cdot \mu \left(1 - \log \sigma\right) \tag{5}$$

is added to the processed node embeddings by the ARMA convolution layers. Thus, the output of the SCG blocks for each feature map includes both a graph, $G = \{\mathcal{F}', \hat{A}\}$, and a residual prediction term, $\hat{y}$.

3.3. Autoregressive Moving Average GNN

In the ARMA GNN stage of our proposed architecture, the formulation of Bianchi et al. [41] was used. Graph convolution can be treated in the graph spectral domain by applying a filter response function, $h(\lambda)$, where λ is an eigenvalue of the Laplacian matrix, $L = D - A$. Graph filtering transforms an input node with a single feature, $\mathcal{F} \in \mathrm{R}^N$, into a filtered node, $\tilde{\mathcal{F}}$ via:

$$\tilde{\mathcal{F}} = U \cdot H_d(\lambda) \cdot U^T \mathcal{F} \tag{6}$$

where U is the eigenvector matrix of L, and $H_d(\lambda)$ is the element-wise application of h on the diagonal matrix of eigenvalues, λ [10].

Due to issues on computational eigendecomposition and non-localized embedding, the polynomial filter, $h_{\text{poly}}(\lambda) = \sum_{k=1}^{K} \omega_k \lambda^k$, was introduced, and the resulting graph filtering relation in Eq.6 is:

$$\tilde{\mathcal{F}} = \sum_{k=0}^{K} \omega_k L^k \mathcal{F} \tag{7}$$

For multi-channeled node feature, the weight, w_k, in Eq. 7 is replaced by a weight matrix, W_k. Several polynomial filters, such as the Chebyshev, were introduced to better model the desired filter responses while minimizing overfitting in the extrapolated frequency range.

As graphs can be interpreted in the spectral domain through the eigenvalues of the Laplacian matrix, several graph convolution operators based on polynomial filters were developed. One example is the Chebyshev polynomial, which replaces the simple L^k with the Chebyshev function, $T_k(L)$ [42]. The graph convolution network (GCN) is a simplification of the Chebyshev spectral filtering with $K = 1$. The autoregressive moving average (ARMA) filter is based on the filter response relation:

$$\tilde{\mathcal{F}} = \left(I + \sum_{k=0}^{K} q_k L^k \right)^{-1} \left(\sum_{k=0}^{K} p_k L^k \right) \mathcal{F} \tag{8}$$

where q_k and p_k are filter coefficients [43]. However, to avoid the intractable inverse matrix operation, the inverse term is approximated by an autoregressive term with additional skip connection. The resulting ARMA convolution is given by the recursion relation:

$$\tilde{\mathcal{F}}^{(t+1)} = \sigma \left(\hat{L} \tilde{\mathcal{F}}^{(t)} W + \mathcal{F} V \right) \tag{9}$$

where $\tilde{\mathcal{F}}^{(t)}$ is the node embedding estimation at iteration t; W and V are learnable weight matrices; $\hat{L}$ is the modified Laplacian; and σ is a nonlinear activation function. It is worth noting that $\tilde{\mathcal{F}}^{(0)} = \mathcal{F}$.

Multiple stacks of $\tilde{\mathcal{F}}^{(t)}$ can be made with ARMA graph convolution. If $t \in \{1, 2, .., T\}$, the set of final node features from J iterations, $\{\tilde{\mathcal{F}}_1^{(T)}\}$ are

Table 1: Details on the architectural design of the proposed model.

Hyper-parameter	$\mathcal{F}_\alpha$	$\mathcal{F}_\beta$	$\mathcal{F}_\gamma$
SCG average pooling output size	(12,12)	(12,12)	(5,5)
SCG number of channels for μ and σ	512	512	1024
Number of channels for each ARMA	128	128	256
Dropout rate for ARMA	0.3	0.3	0.3
Activation function used in ARMA	ReLU	ReLU	ReLU
Number of channels for MLP layer	128	128	256
Weighted concatenation coefficient	$\alpha = 0.25$	$\beta = 0.5$	-

averaged to yield the output, $\tilde{\mathcal{F}}$. The GNN layers in our proposed model uses a special case of the ARMA convolution with $T = 3$, and $J = 1$. This reduces to a single-branch ARMA filtering layer. Referring to equations 5 and 9, the ARMA filtering layer used in our proposed model can be mathematically expressed as:

$$\tilde{\mathcal{F}} = \left(\tilde{\mathcal{F}}^{(3)} \circ \tilde{\mathcal{F}}^{(2)} \circ \tilde{\mathcal{F}}^{(1)}\right)(\mathcal{F}) + \rho\hat{y} \tag{10}$$

where $\rho = 1$ if $\tilde{\mathcal{F}}$ is from the final ARMA convolutional layer and $\rho = 0$ otherwise. Two ARMA GNN layers are used in each feature map processing stage.

3.4. Feature Aggregation and Classification Head

The output node features from each feature processing level are subjected to an average pooling to yield a single feature vector. Feature vectors h_α, h_β, and h_γ correspond to the pooled node features following the GNN processing of F_α, F_β, and F_γ, respectively. Each feature vector is passed on a single fully-connected layer (or MLP) for feature postprocessing. The feature vectors from the three GNN processing levels are combined through weighted concatenation:

$$h' = \text{Concat}\left[\frac{\alpha}{\alpha+\beta+1}MLP(h_\alpha), \frac{\beta}{\alpha+\beta+1}MLP(h_\beta), MLP(h_\gamma)\right] \tag{11}$$

Ultimately, the concatenated feature vector, h', proceeds to the output softmax layer to yield the prediction score per class.

Table 2: General information on some of the selected MedMNIST 2D dataset for classification model benchmarking [34].

Dataset	Image Modality	Number of classes	Number of train/val/test split
DermaMNIST	Dermatoscope	7	7,007 / 1,003 / 2,005
OCTMNIST	Retinal OCT	4	97,477 / 10,832 / 1,000
PneumoniaMNIST	Chest x-ray	2	4,708 / 524 / 624
BreastMNIST	Breast ultrasound	2	546 / 78 / 156
BloodMNIST	Blood cell microscope	8	11,959 / 1,712 / 3,421
OrganAMNIST	Abdominal CT	11	34,561 / 6,491 / 17,778
OrganCtMNIST	Abdominal CT	11	546 / 78 / 156
OrganSMNIST	Abdominal CT	11	546 / 78 / 156

4. Experimental Results

4.1. Architecture and Training Implementation

A summary of the standard architecture is shown in Table 1. To evaluate our proposed model in medical imaging, we used four 2D imaging datasets from the MedMNIST V2 repository [34]. The datasets used on our experiments are defined in Table 2. MedMNIST provides an excellent platform for evaluating classification models in medical imaging due to its diverse range of modalities. DermaMNIST involves a classification of several types of pigmented skin lesion from dermatoscope images. PneumoniaMNIST, OCTMNIST, and BreastMNIST are classification tasks for detection of a presence of a disease corresponding to pediatric x-ray, retinal optical coherence tomography (OCT) and breast ultrasound, respectively. Normal peripheral blood classification using cell microscopy was also included as a benchmarking model. Lastly, datasets on organ identification in abdominal computed tomography (CT) at different body planes are used for model performance benchmarking.

Although MedMNIST comes with different image dimensions, we limit our experiment on the 224×224 since they closely matched the dimensions utilized in real clinical environments. The images in all datasets underwent z-normalization prior to being fed to the classification models. All models implemented on this study were trained and tested on the same set of training and testing images, respectively, for all datasets. Our proposed model was trained using the adaptive moment estimation (Adam) optimizer [44] for

Table 3: Comparison of the performance of IncARMAG with various published models that were evaluated on DermaMNIST, OCTMNIST, PneumoniaMNIST, and BreastMNIST datasets.

Model	DermaMNIST		OCTMNIST		Pneumonia MNIST		BreastMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
ResNet18 [34]	0.920	0.754	0.958	0.763	0.956	0.864	0.891	0.833
ResNet50 [34]	0.912	0.731	0.958	0.776	0.962	0.884	0.857	0.812
auto-sklearn [34]	0.902	0.719	0.887	0.601	0.942	0.855	0.836	0.803
AutoKeras [34]	0.915	0.749	0.955	0.763	0.947	0.878	0.871	0.831
Google AutoML Vision [34]	0.914	0.768	0.963	0.771	0.991	0.946	0.919	0.861
MedViT-T [33]	0.914	0.768	0.961	0.767	0.993	0.949	0.934	**0.896**
MedViT-S [33]	0.937	0.780	0.960	0.782	**0.995**	**0.961**	**0.938**	**0.897**
MedViT-L [33]	0.920	0.773	0.945	0.761	0.991	0.921	0.929	0.883
EHDFL [45]	0.917	0.769	0.917	0.769	0.968	0.883	0.894	0.897
IncARMAG (Ours)	**0.936**	**0.802**	**0.985**	**0.873**	0.938	0.885	0.916	0.872

100 epochs with an initial learning rate of 1E-04 that decays by a factor of 0.1 for every epoch milestones 50 and 75. A batch size of 16 was used for all implemented models and for all datasets in this study. The loss function used for all the training instances was the categorical cross entropy function. A regularization decay factor of 0.0001 was used for all models except for the transformer-based models in order to avoid nonfinite loss. Data augmentation was not employed during the training phase. All implementations were carried out using PyTorch and Torchvision in Nvidia A100 GPU.

4.2. Comparison with SOTA Models

Tables 3 and 4 present the accuracy (ACC) and area under the receiver operating curve (AUC) of our proposed IncARMAG across eight MedMNIST datasets. We cite the results of some studies that has carried out experiments on MedMNIST datasets for comparison. These include ResNets and autoML models from the original MedMNIST study [34], MedViT [33], and evolutionary hybrid domain feature learning (EHDFL) [45]. IncARMAG achieved the highest accuray and AUC on DermaMNIST, OCTMNIST, BloodMNIST, and OrganCMNIST, whereas MedViT model performance dominated at the PneumoniaMNIST, BreastMNIST, and OrganSMNIST. An accuracy

Table 4: Comparison of the performance of IncARMAG with various published models that were evaluated on BloodMNIST, OrganAMNIST, OrganCMNIST, and OrganSMNIST datasets.

Model	BloodMNIST		OrganAMNIST		OrganCMNIST		OrganSMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
ResNet18 [34]	0.998	0.958	0.997	0.935	0.994	0.920	0.974	0.782
ResNet50 [34]	0.997	0.950	**0.998**	**0.947**	0.993	0.911	0.975	0.785
auto-sklearn [34]	0.984	0.878	0.963	0.762	0.976	0.829	0.945	0.672
AutoKeras [34]	0.998	0.961	0.994	0.905	0.990	0.879	0.974	**0.813**
Google AutoML Vision [34]	0.998	0.966	0.990	0.886	0.988	0.877	0.964	0.749
MedViT-T [33]	0.996	0.950	0.995	0.931	0.991	0.901	0.972	0.789
MedViT-S [33]	0.997	0.951	0.996	0.928	0.993	0.916	**0.987**	0.805
MedViT-L [33]	0.996	0.954	0.997	0.943	0.994	0.922	0.973	0.806
EHDFL [45]	-	-	0.997	0.936	0.994	0.916	0.974	0.784
IncARMAG (Ours)	**0.999**	**0.987**	0.993	0.927	**0.995**	**0.934**	0.975	0.801

improvement greater than 2 % is given by our proposed IncARMAG for DermaMNIST, OCTMNIST, and BloodMNIST. Significant increase in accuracy is observed on the OCTMNIST dataset evaluation. However, performance decline is noted in BreastMNIST, PneumoniaMNIST, and OrganAMNIST when comparing IncARMAG with the benchmark models. For OrganAMNIST, ResNet50 achieved the highest accuracy and AUC. The architecture of our IncARMAG potentially matches the generalizability of MedViT in medical image classification based on the ranking statistics of performance in Tables 3 and 4. Despite not using any data augmentation techniques, our proposed model is still capable of generating reliable predictions following model training, with the exception of pediatric pneumonia classification task in PneumoniaMNIST. IncARMAG has also surpassed the performance of autoML models and ResNet architectures in most of the MedMNIST datasets.

To further assess the capabilities of our model, we trained and evaluated several CNN, transformer, and hybrid-type models on four MedMNIST datasets using the aforementioned data preprocessing and training hyperparameters in Section 4.1. These models are the ResNet series [14], Inception V3 [38], EfficientNet V2 [15], ConvNext-B [13], ViT-B [46], MaxViT [47], Swin-S [48], DEIT [49], RIFormer [50], HorNet [51], and ViG [26]. These models were implemented using torchvision and mmpretrain libraries [52].

Table 5: Performance of IncARMAG and benchmarking models on DermaMNIST and PneumoniaMNIST datasets.

Model	DermaMNIST			PneumoniaMNIST		
	Acc	Prec	Rec.	Acc.	Prec.	Rec.
ResNet101	0.757	0.575	0.569	0.865	0.826	0.995
ResNet152	0.783	0.628	0.604	0.859	0.819	0.995
Inception V3	0.782	0.587	0.609	0.864	0.822	0.997
EfficientNet V2	0.764	0.597	0.526	0.852	0.813	0.995
ConvNext	0.756	0.583	0.522	0.872	0.834	0.992
ViT-B	0.730	0.493	0.480	0.830	0.794	0.977
MaxViT	0.773	0.629	0.524	0.851	0.811	0.992
Swin-S	0.782	0.587	0.609	0.625	0.625	**1.00**
DEIT V3	0.732	0.514	0.509	0.835	0.800	0.982
RIFormer	0.730	0.485	0.430	0.825	0.791	0.980
HorNet	0.758	0.565	0.574	0.830	0.806	0.982
ViG	0.749	0.506	0.491	0.870	0.832	0.992
IncARMAG (Ours)	**0.795**	**0.667**	**0.617**	**0.886**	**0.849**	0.995

Overall accuracy, macro recall, and macro precision are reported for each model. Overall accuracy offers a general insight on the model performance across the evaluation dataset while macro precision and recall highlight false positive and false negative predictions, respectively, with less sensitivity to class imbalances [53].

The experimental results of our proposed and benchmarking models for the four selected MedMNIST datasets are shown in Tables 5 and 6, which shows that IncARMAG outperforms the counterpart benchmarking models in terms of accuracy and precision. ResNet and Inception have consistently attained above-median performance for the four datasets. The performance of transformer-based models varies across the datasets, which can be attributed to the slower convergence as a function of training data points [46]. The motivation behind the SCG in IncARMAG and attention mechanism in transformer-based models partially share some resemblance, which is set to allow learning long-range dependencies among the nodes for graphs or tokens for transformers. Thus, our proposed model leverages the similarity with attention mechanism while maintaining faster performance convergence as demonstrated by CNN models. Although both IncARMAG and ViG make use of GNN approach in their architecture, there are some significant discrep-

Table 6: Performance of IncARMAG and benchmarking models on BloodMNIST and BreastMNIST datasets.

Model	BloodMNIST			BreastMNIST		
	Acc	Prec	Rec.	Acc.	Prec.	Rec.
ResNet101	0.985	0.984	0.986	0.814	0.876	0.868
ResNet152	0.985	0.984	0.986	0.821	**0.898**	0.851
Inception V3	0.984	0.985	0.986	0.840	0.908	0.868
EfficientNet V2	0.984	0.984	0.984	0.731	0.731	**1.0**
ConvNext	0.981	0.983	0.982	0.808	0.818	0.948
ViT-B	0.956	0.952	0.0.949	0.711	0.805	0.798
MaxViT	0.983	0.984	0.0.984	0.827	0.822	0.974
Swin-S	0.889	0.870	0.0.856	0.673	0.756	0.816
DEIT V3	0.958	0.960	0.954	0.750	0.800	0.877
RIFormer	0.949	0.960	0.953	0.699	0.734	0.921
HorNet	0.851	0.869	0.808	0.737	0.739	0.991
ViG	0.983	0.984	0.984	0.808	0.862	0.877
IncARMAG (Ours)	**0.987**	**0.987**	0.986	**0.872**	**0.898**	0.930

ancies in accuracy on two out of four of the benchmark datasets.

Table 5 shows that although the overall accuracy of the models evaluated under the DermaMNIST dataset are reasonable in value, the precision and recall per class reveals unacceptable performance. These results entail that model degeneracy and/or imbalanced class training may have occurred. Insights on the error distribution can be visualized through the confusion matrices shown in Figs. 3a-3d for the four datasets. Fig. 3a shows the IncARMAG has a strong accuracy owing to the slight prediction bias towards the prominent melanoma label in an imbalanced dataset.

There is minimal to no improvement in classification accuracy compared with some of the benchmark models for PneumoniaMNIST and BloodMNIST experiments. The confusion matrix of IncARMAG on BloodMNIST experiment shows strong classification accuracy across all labels, as illustrated in Fig. 3c. A large gap in accuracy can be found between IncARMAG and the benchmark models in the BreastMNIST dataset. Swin-S was able to surpass IncARMAG in terms of recall as this model gives a completely degenerate prediction that leads to nonexistent false negative.

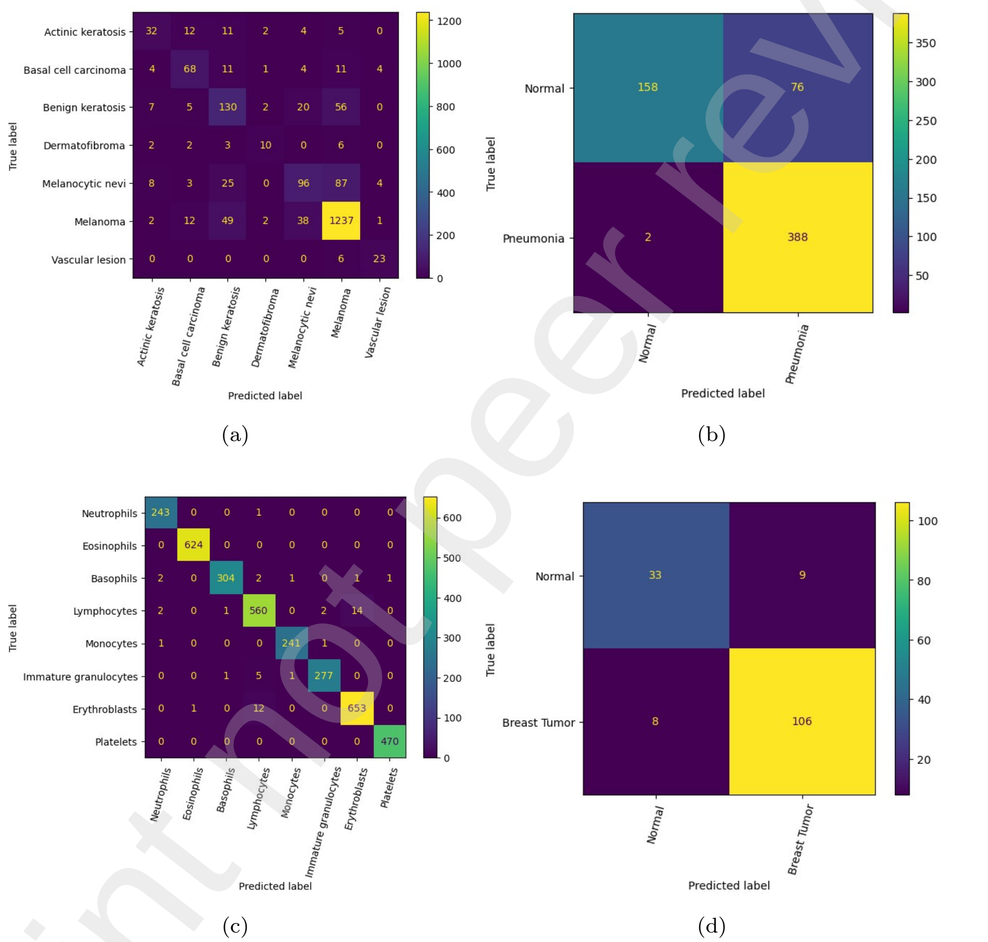

Figure 3: Confusion matrix of IncARMAG predictions on the test set of (a) DermaMNIST, (b) PneumoniaMNIST, (c) BloodMNIST, and (d) BreastMNIST.

4.3. Enhancing Performance through Multi-Level GNNs

We have also included two additional variants of our proposed model. One of the IncARMAG modified architecture, denoted as IncARMAG v2, has a single GNN-based feature processing at the last feature map, F_γ. We have also explored varying the feature map levels, F_α and F_β, that were connected to the GNN-based feature refinement. IncARMAG V3 has F_α and F_β equal to the feature maps at Inception blocks 6 and 7, respectively. The evaluation results of these IncARMAG variants on the four MedMNIST datasets in Table 2 are shown in Figs. 4.

There is a clear benefit in using multi-level feature processing framework on the DermaMNIST, PneumoniaMNIST, and BreastMNIST datasets. Adding auxiliary classifiers may result in an increased accuracy, which can be ascribed to the feature refinement and/or inherent regularization as argued in the Inception V3 paper [38]. Comparison of results of Inception V3 on Tables 3-4 with Figs 4a and 4b shows how we are able to increase the predictive performance from the classical Inception V3 by considering ARMA convolutional layers as the multi-level auxiliary classifiers. However, there is no significant difference in accuracy between IncARMAG and IncARMAG V3 on all four datasets. Thus, the benefit of auxiliary GNN classifiers in IncARMAG may be considered more of as an added regularizer while encouraging global feature diversity through the SCG and ARMA modules.

5. Discussion

One way to determine if meaningful representations from images are learned by IncARMAG is via visualization of feature activation heatmaps. In Fig. 5, we show a comparison of feature activations of ViG and IncARMAG obtained on several images using gradient-weighted class activation mapping (grad-cam). Grad-cam initially calculates class-specific weight importance, which are incorporated to the weighted-sum of feature activation maps [54].

The first row illustrates how the ViG model was able capture features localized in the melanoma region for prediction. On the other hand, IncARMAG has better coverage of the region-of-interest while having partial inclusion of areas outside the melanoma region. Superior localization of feature activation importance is demonstrated by IncARMAG compared with ViG for the PneumoniaMNIST image prediction. In this case, IncARMAG obtained hotspots on lung areas with blurry infiltrates. Both IncARMAG and Vision-G have strengths and weaknesses in terms of the extent and precision

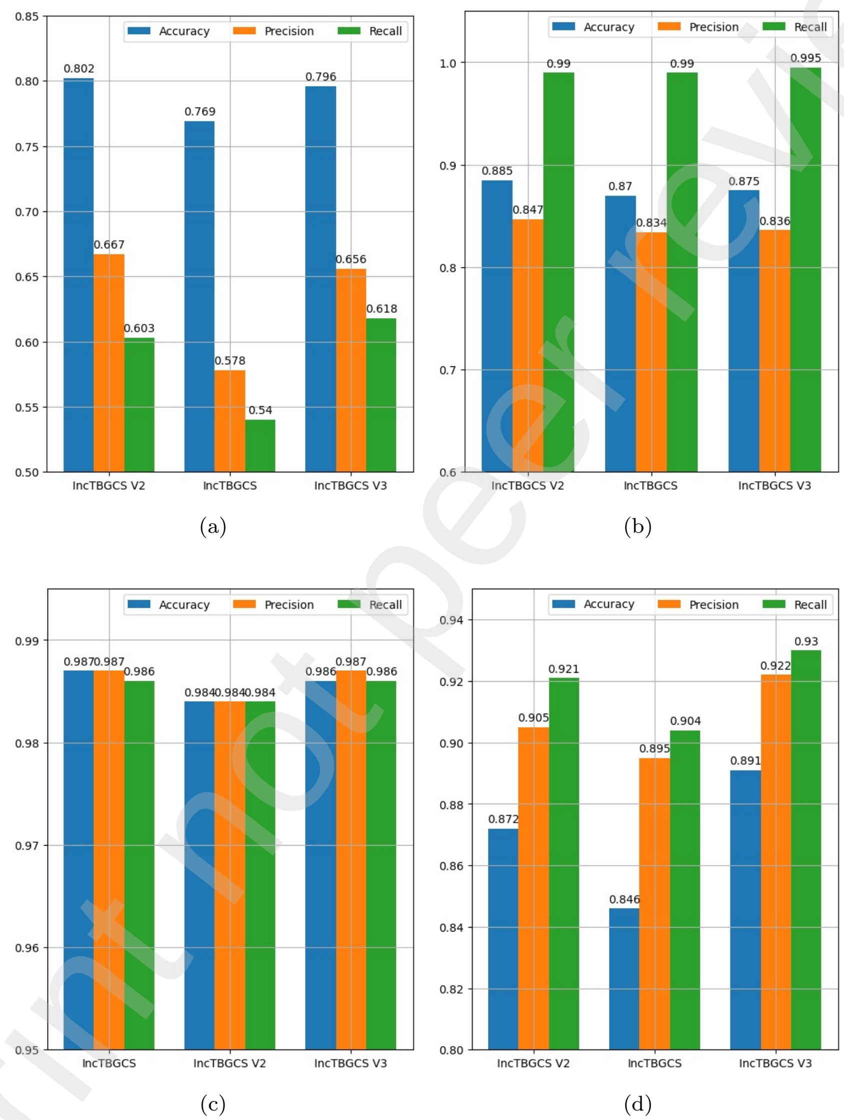

Figure 4: Performance metrics of various IncARMAG variants evaluated on the test set of (a) DermaMNIST, (b) PneumoniaMNIST, (c) BloodMNIST, and (d) BreastMNIST.

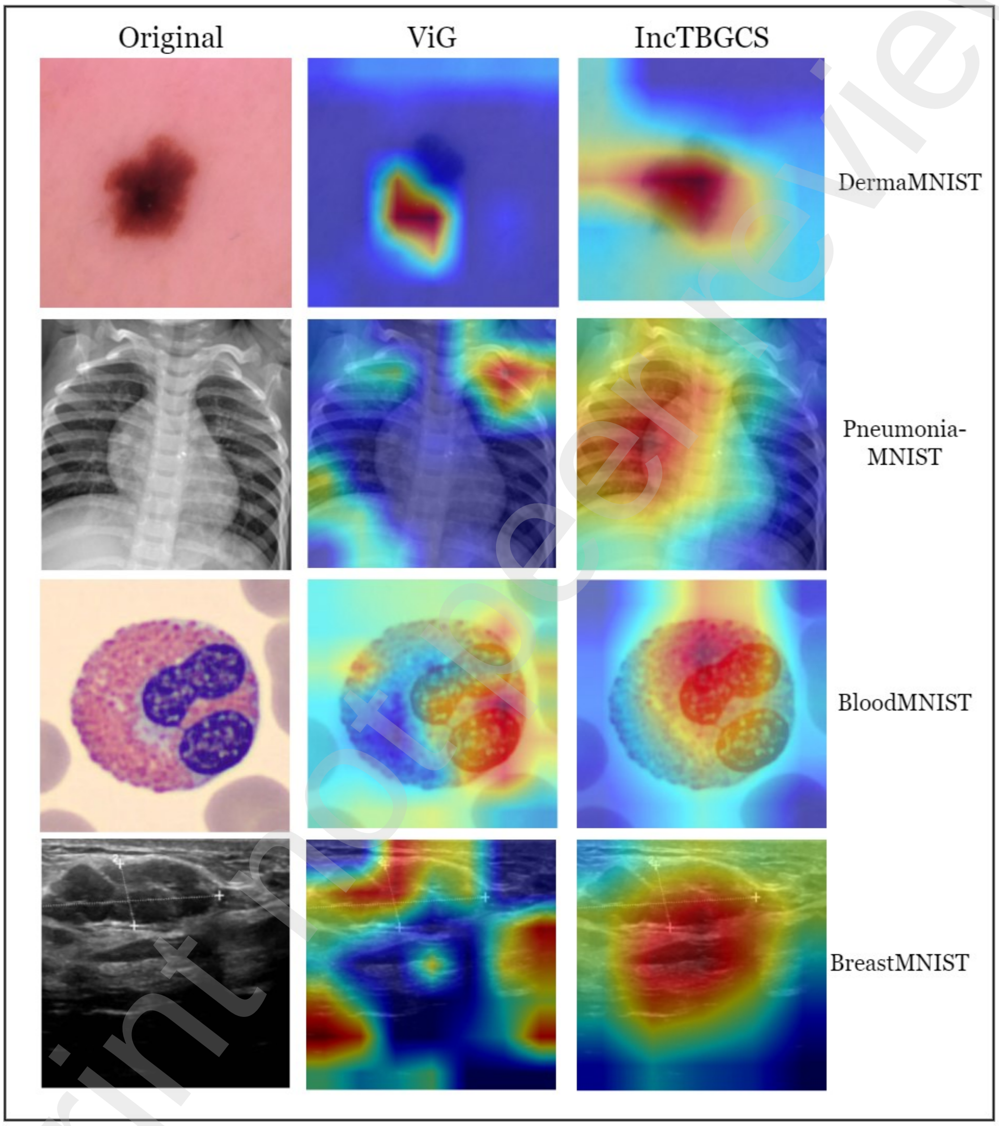

Figure 5: Grad-cam visualization of ViG and IncARMAG feature activations on a sample image in each dataset. Red and blue colors in the heatmap correspond to the maximum and minimum extremes of activation values, respectively.

of feature activation coverage in the eosinophil image. Lastly, IncARMAG form one aggregated hotspot with significant extension beyond the breast tumor region, which is partially justified for distinguishing the ROI from the background structures. Overall, IncARMAG has shown wide feature activation coverage in class score prediction with hotspots on the regions-of-interest.

It is imperative to develop classification models with great accuracy given the gravity of errors within the medical domain. This study has enhanced the foundational Inception model for medical image classification. For future work, additional data preprocessing techniques, such as data augmentation, should be considered to enhance the IncARMAG model performance. In addition, hyper-parameter optimization can be utilized to determine the optimal model hyper-parameters, including the SCG and GNN hidden dimensions, number of iterations in each ARMA layer, etc. Although the classification models we have reported require further improvement on some of the datasets, IncARMAG provides a potential direction on how to develop computer-aided classification in medical images using multi-level GNN for feature processing. IncARMAG can also be used as a groundwork for developing computer vision models on other medical imaging tasks, such as image segmentation and registration. For example, IncARMAG can serve as a backbone encoder network in an encoder-decoder segmentation architecture.

Translating these AI models to clinical utility requires substantial validation from prospective and retrospective studies. An initial challenge is to increase the number of samples and cases in order to improve accuracy and generalizability of classification models such that reliable predictions can be generated with unseen data [55]. Ensuring robust performance also requires addressing algorithmic biases and data drift. Once a sufficient accuracy has been achieved, conducting prospective studies for clinical validation further strengthens the practical effectiveness of AI models in clinical settings [56].

6. Conclusion

The novelty of our proposed IncARMAG lies on the adoption of SCG-ARMA GNN framework on different hierarchical feature levels to capture long-range representations. Our experiments has demonstrated that IncARMAG ranks as one of the top models in classification of medical images that spans different imaging modalities. IncARMAG has also outperformed various CNN and transformer-based state-of-the-art models based on more extensive experiments on four medical imaging datasets. The boost in performance with multi-level version of IncARMAG has presented the advantages of using additional SCG-GNN as auxiliary heads, which can be attributed to the regularization and/or feature diversity.

Conflict of interest

The author declares that they have no conflict of interest.

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References

[1] S. Hussain, I. Mubeen, N. Ullah, S. S. U. D. Shah, B. A. Khan, M. Zahoor, R. Ullah, F. A. Khan, M. A. Sultan, et al., Modern diagnostic imaging technique applications and risk factors in the medical field: a review, BioMed research international 2022 (2022).

[2] P. P. Dendy, B. Heaton, Physics for diagnostic radiology, CRC press, 2011.

[3] J. Potočnik, S. Foley, E. Thomas, Current and potential applications of artificial intelligence in medical imaging practice: A narrative review, Journal of Medical Imaging and Radiation Sciences (2023). doi:https://doi.org/10.1016/j.jmir.2023.03.033.

[4] Q. Tang, Y. Liu, H. Liu, Medical image classification via multiscale representation learning, Artificial intelligence in medicine 79 (2017) 71–78.

[5] R. Miotto, F. Wang, S. Wang, X. Jiang, J. T. Dudley, Deep learning for healthcare: review, opportunities and challenges, Briefings in bioinformatics 19 (6) (2018) 1236–1246.

[6] J.-G. Lee, S. Jun, Y.-W. Cho, H. Lee, G. B. Kim, J. B. Seo, N. Kim, Deep learning in medical imaging: general overview, Korean journal of radiology 18 (4) (2017) 570.

[7] J. J. Van Griethuysen, A. Fedorov, C. Parmar, A. Hosny, N. Aucoin, V. Narayan, R. G. Beets-Tan, J.-C. Fillion-Robin, S. Pieper, H. J. Aerts, Computational radiomics system to decode the radiographic phenotype, Cancer research 77 (21) (2017) e104–e107.

[8] M. W. Wagner, K. Namdar, A. Biswas, S. Monah, F. Khalvati, B. B. Ertl-Wagner, Radiomics, machine learning, and artificial intelligence—what the neuroradiologist needs to know, Neuroradiology (2021) 1–11.

[9] A. Shah, M. Shah, A. Pandya, R. Sushra, R. Sushra, M. Mehta, K. Patel, K. Patel, A comprehensive study on skin cancer detection using artificial neural network (ann) and convolutional neural network (cnn), Clinical eHealth (2023).

[10] Y. Ma, J. Tang, Deep learning on graphs, Cambridge University Press, 2021.

[11] A. Krizhevsky, I. Sutskever, G. E. Hinton, Imagenet classification with deep convolutional neural networks, Advances in neural information processing systems 25 (2012).

[12] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, arXiv (2014). doi:https://doi.org/10.48550/arXiv.1409.1556.

[13] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, S. Xie, A convnet for the 2020s, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022, pp. 11976–11986.

[14] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 770–778.

[15] M. Tan, Q. Le, Efficientnetv2: Smaller models and faster training, in: International conference on machine learning, PMLR, 2021, pp. 10096–10106.

[16] B. Zoph, Q. V. Le, Neural architecture search with reinforcement learning, arXiv preprint arXiv:1611.01578 (2016). doi:https://doi.org/10.48550/arXiv.1611.01578.

[17] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, et al., Searching for mobilenetv3, in: Proceedings of the IEEE/CVF international conference on computer vision, 2019, pp. 1314–1324.

[18] C.-Y. Wang, H.-Y. M. Liao, Y.-H. Wu, P.-Y. Chen, J.-W. Hsieh, I.-H. Yeh, Cspnet: A new backbone that can enhance learning capability of cnn, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops, 2020, pp. 390–391.

[19] J. Wang, K. Sun, T. Cheng, B. Jiang, C. Deng, Y. Zhao, D. Liu, Y. Mu, M. Tan, X. Wang, et al., Deep high-resolution representation learning for visual recognition, IEEE transactions on pattern analysis and machine intelligence 43 (10) (2020) 3349–3364. doi:10.1109/TPAMI.2020.2983686.

[20] V. Vasudevan, M. Bassenne, M. T. Islam, L. Xing, Image classification using graph neural network and multiscale wavelet superpixels, Pattern Recognition Letters 166 (2023) 89–96. doi:https://doi.org/10.1016/j.patrec.2023.01.003.

[21] M. Krzywda, S. Łukasik, A. H. Gandomi, Graph neural networks in computer vision-architectures, datasets and common approaches, in: 2022 International Joint Conference on Neural Networks (IJCNN), IEEE, 2022, pp. 1–10. doi:10.1109/IJCNN55064.2022.9892658.

[22] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Lio, Y. Bengio, et al., Graph attention networks, stat 1050 (20) (2017) 10–48550.

[23] F. Monti, D. Boscaini, J. Masci, E. Rodola, J. Svoboda, M. M. Bronstein, Geometric deep learning on graphs and manifolds using mixture model cnns, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2017, pp. 5115–5124.

[24] P. H. Avelar, A. R. Tavares, T. L. da Silveira, C. R. Jung, L. C. Lamb, Superpixel image classification with graph attention networks, in: 2020 33rd SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI), IEEE, 2020, pp. 203–209. doi:10.1109/SIBGRAPI51738.2020.00035.

[25] D. Yao, Z. Zhi-li, Z. Xiao-feng, C. Wei, H. Fang, C. Yao-ming, W.-W. Cai, Deep hybrid: multi-graph neural network collaboration for hyperspectral image classification, Defence Technology 23 (2023) 164–176. doi:https://doi.org/10.1016/j.dt.2022.02.007.

[26] K. Han, Y. Wang, J. Guo, Y. Tang, E. Wu, Vision gnn: An image is worth graph of nodes, Advances in neural information processing systems 35 (2022) 8291–8303.

[27] Z. Fei, J. Guo, H. Gong, L. Ye, E. Attahi, B. Huang, A gnn architecture with local and global-attention feature for image classification, IEEE Access (2023). doi:10.1109/ACCESS.2023.3285246.

[28] Y. R. Park, Y. J. Kim, W. Ju, K. Nam, S. Kim, K. G. Kim, Comparison of machine and deep learning for the classification of cervical cancer based on cervicography images, Scientific Reports 11 (1) (2021) 16143. doi:https://doi.org/10.1038/s41598-021-95748-3.

[29] S. De, R. J. Stanley, C. Lu, R. Long, S. Antani, G. Thoma, R. Zuna, A fusion-based approach for uterine cervical cancer histology image classification, Computerized medical imaging and graphics 37 (7-8) (2013) 475–487. doi:https://doi.org/10.1016/j.compmedimag.2013.08.001.

[30] Z. Zhuang, Z. Yang, A. N. J. Raj, C. Wei, P. Jin, S. Zhuang, Breast ultrasound tumor image classification using image decomposition and fusion based on adaptive multi-model spatial feature fusion, Computer methods and programs in biomedicine 208 (2021) 106221. doi:https://doi.org/10.1016/j.cmpb.2021.106221.

[31] D. M. Ibrahim, N. M. Elshennawy, A. M. Sarhan, Deep-chest: Multi-classification deep learning model for diagnosing covid-19, pneumonia, and lung cancer chest diseases, Computers in biology and medicine 132 (2021) 104348. doi:https://doi.org/10.1016/j.compbiomed.2021.104348.

[32] S. B. Yengec-Tasdemir, Z. Aydin, E. Akay, S. Dogan, B. Yilmaz, An effective colorectal polyp classification for histopathological images based on supervised contrastive learning, Computers in Biology and Medicine 172 (2024) 108267. doi:https://doi.org/10.1016/j.compbiomed.2024.108267.

[33] O. N. Manzari, H. Ahmadabadi, H. Kashiani, S. B. Shokouhi, A. Ayatollahi, Medvit: a robust vision transformer for generalized medical image classification, Computers in Biology and Medicine 157 (2023) 106791. doi:https://doi.org/10.1016/j.compbiomed.2023.106791.

[34] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, B. Ni, Medmnist v2-a large-scale lightweight benchmark for 2d and 3d biomedical image classification, Scientific Data 10 (1) (2023) 41. doi:https://doi.org/10.1038/s41597-022-01721-8.

[35] Y. Yue, Z. Li, Medmamba: Vision mamba for medical image classification, arXiv preprint arXiv:2403.03849 (2024). doi:https://doi.org/10.48550/arXiv.2403.03849.

[36] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, A. Rabinovich, Going deeper with convolutions, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2015, pp. 1–9. doi:10.1109/CVPR.2015.7298594.

[37] X. Ding, X. Zhang, J. Han, G. Ding, Scaling up your kernels to 31x31: Revisiting large kernel design in cnns, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022, pp. 11963–11975. doi:10.1109/CVPR52688.2022.01166.

[38] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 2818–2826.

[39] Q. Liu, M. Kampffmeyer, R. Jenssen, et al., Self-constructing graph convolutional networks for semantic labeling, in: IGARSS 2020-2020 IEEE International Geoscience and Remote Sensing Symposium, IEEE, 2020, pp. 1801–1804. doi:10.1109/IGARSS39084.2020.9324719.

[40] T. N. Kipf, M. Welling, Variational graph auto-encoders, arXiv preprint arXiv:1611.07308 (2016). doi:https://doi.org/10.48550/arXiv.1611.07308.

[41] F. M. Bianchi, D. Grattarola, L. Livi, C. Alippi, Graph neural networks with convolutional arma filters, IEEE transactions on

pattern analysis and machine intelligence 44 (7) (2021) 3496–3507. doi:10.1109/TPAMI.2021.3054830.

[42] M. Defferrard, X. Bresson, P. Vandergheynst, Convolutional neural networks on graphs with fast localized spectral filtering, Advances in neural information processing systems 29 (2016).

[43] N. Tremblay, P. Gonçalves, P. Borgnat, Design of graph filters and filterbanks, in: Cooperative and Graph Signal Processing, Elsevier, 2018, pp. 299–324.

[44] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, arXiv preprint arXiv:1412.6980 (2014). doi:https://doi.org/10.48550/arXiv.1412.6980.

[45] Q. Han, M. Hou, H. Wang, C. Wu, S. Tian, Z. Qiu, B. Zhou, Ehdfl: Evolutionary hybrid domain feature learning based on windowed fast fourier convolution pyramid for medical image classification, Computers in Biology and Medicine 152 (2023) 106353. doi:https://doi.org/10.1016/j.compbiomed.2022.106353.

[46] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, arXiv preprint arXiv:2010.11929 (2020). doi:https://doi.org/10.48550/arXiv.2010.11929.

[47] Z. Tu, H. Talebi, H. Zhang, F. Yang, P. Milanfar, A. Bovik, Y. Li, Maxvit: Multi-axis vision transformer, in: European conference on computer vision, Springer, 2022, pp. 459–479. doi:https://doi.org/10.1007/978-3-031-20053-327.

[48] Z. Liu, H. Hu, Y. Lin, Z. Yao, Z. Xie, Y. Wei, J. Ning, Y. Cao, Z. Zhang, L. Dong, et al., Swin transformer v2: Scaling up capacity and resolution, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022, pp. 12009–12019. doi:10.1109/CVPR52688.2022.01170.

[49] H. Touvron, M. Cord, H. Jego, Deit iii: Revenge of the vit, in: European conference on computer vision, 2022, pp. 516–533.

[50] J. Wang, S. Zhang, Y. Liu, T. Wu, Y. Yang, X. Liu, K. Chen, P. Luo, D. Lin, Riformer: Keep your vision backbone effective but removing token mixer, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 14443–14452. doi:10.1109/CVPR52729.2023.01388.

[51] Y. Rao, W. Zhao, Y. Tang, J. Zhou, S.-N. Lim, J. Lu, Hornet: Efficient high-order spatial interactions with recursive gated convolutions, Advances in Neural Information Processing Systems 35 (2022) 10353–10366.

[52] M. Contributors, Openmmlab's pre-training toolbox and benchmark (2023).

[53] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al., Scikit-learn: Machine learning in python, the Journal of machine Learning research 12 (2011) 2825–2830.

[54] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, in: Proceedings of the IEEE international conference on computer vision, 2017, pp. 618–626. doi:10.1109/ICCV.2017.74.

[55] R. J. Ellis, R. M. Sander, A. Limon, Twelve key challenges in medical machine learning and solutions (2022). doi:https://doi.org/10.1016/j.ibmed.2022.100068.

[56] E. H. Weissler, T. Naumann, T. Andersson, R. Ranganath, O. Elemento, Y. Luo, D. F. Freitag, J. Benoit, M. C. Hughes, F. Khan, et al., The role of machine learning in clinical research: transforming the future of evidence generation, Trials 22 (2021) 1–15. doi:https://doi.org/10.1186/s13063-021-05489-x.